

1 Covariance

1.1 Definition: Autocovariance

对于同一 time series x_t 内两点 s 和 t , 定义其之间的 covariance 为 **autocovariance** γ_x :

$$\gamma_x(s, t) = \text{Cov}(x_s, x_t) = \mathbb{E}[(x_s - \mu_s)(x_t - \mu_t)]$$

当 $s = t$ 时, 有:

$$\gamma_x(t, t) = \mathbb{E}[(x_t - \mu_t)^2] = \text{Var}[x_t]$$

Example: White Noise 的 Autocovariance

对于 white noise w_t , 不同 time points 之间没有 dependence, 因此有

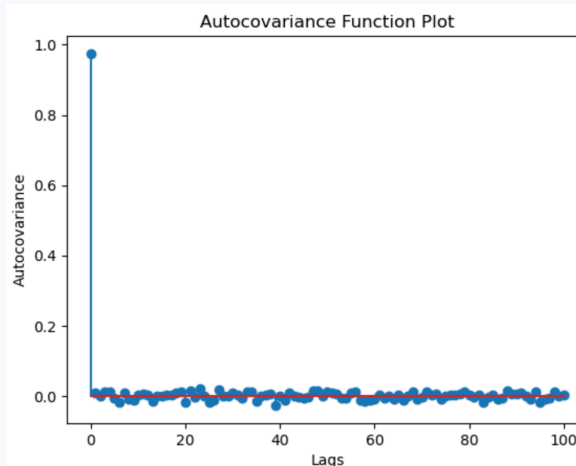
$$\gamma_w(s, t) = \text{Cov}(w_s, w_t) = \begin{cases} \sigma_w^2, & \text{if } s = t \\ 0, & \text{if } s \neq t \end{cases}$$

我们可以使用 `statsmodels` 包中的 `statsmodels.tsa.stattools.acovf` 函数来估计 autocovariance (具体计算方法会在稍后介绍):



Python

```
1 from statsmodels.tsa.stattools import acovf
2
3 # Autocovariance of white noise
4 nt = 10000
5 w = white_noise(nt)
6 acovf_values = acovf(w, nlag=100)
7
8 # The acovf doesn't have a separate plotting function, so we
9 # will just use a stem plot.
10 plt.stem(acovf_values)
11 plt.title('Autocovariance Function Plot')
12 plt.xlabel('Lags')
13 plt.ylabel('Autocovariance')
14 plt.show();
```



1.2 Covariance of Linear Combination

对于 random variables $U = \sum_{j=1}^m a_j X_j$ 和 $V = \sum_{k=1}^r b_k Y_k$, 其 covariance 为:

$$\text{Cov}(U, V) = \sum_{j=1}^m \sum_{k=1}^r a_j b_k \text{Cov}(X_j, Y_k)$$

且有:

$$\text{Var}[U] = \text{Cov}(U, U) \quad \text{Var}[V] = \text{Cov}(V, V)$$

Example: 3-point Moving Average 的 Autocovariance

对于 3-point moving average v_t , 有:

$$\begin{aligned} \gamma_v(s, t) &= \text{Cov}(v_s, v_t) = \text{Cov}\left(\frac{1}{3}(w_{s-1} + w_s + w_{s+1}), \frac{1}{3}(w_{t-1} + w_t + w_{t+1})\right) \\ &= \frac{1}{9} \text{Cov}(w_{s-1} + w_s + w_{s+1}, w_{t-1} + w_t + w_{t+1}) \end{aligned}$$

若 $s = t$, 有:

$$\begin{aligned} \gamma_v(t, t) &= \text{Cov}(v_t, v_t) = \frac{1}{9} \text{Cov}(w_{t-1} + w_t + w_{t+1}, w_{t-1} + w_t + w_{t+1}) \\ &= \frac{1}{9} (\text{Cov}(w_{t-1}, w_{t-1}) + \text{Cov}(w_t, w_t) + \text{Cov}(w_{t+1}, w_{t+1}) \\ &\quad + 2\text{Cov}(w_{t-1}, w_t) + 2\text{Cov}(w_t, w_{t+1}) + 2\text{Cov}(w_{t-1}, w_{t+1})) \\ &= \frac{1}{9} (\sigma_w^2 + \sigma_w^2 + \sigma_w^2 + 0 + 0 + 0) \\ &= \frac{1}{3} \sigma_w^2 \end{aligned}$$

若 $s = t + 1$, 有:

$$\begin{aligned} \gamma_v(t+1, t) &= \text{Cov}(v_{t+1}, v_t) = \frac{1}{9} \text{Cov}(w_t + w_{t+1} + w_{t+2}, w_{t-1} + w_t + w_{t+1}) \\ &= \frac{1}{9} (\text{Cov}(w_t, w_t) + \text{Cov}(w_{t+1}, w_{t+1})) \\ &= \frac{2}{9} \sigma_w^2 \end{aligned}$$

若考虑更多 lags, 则有:

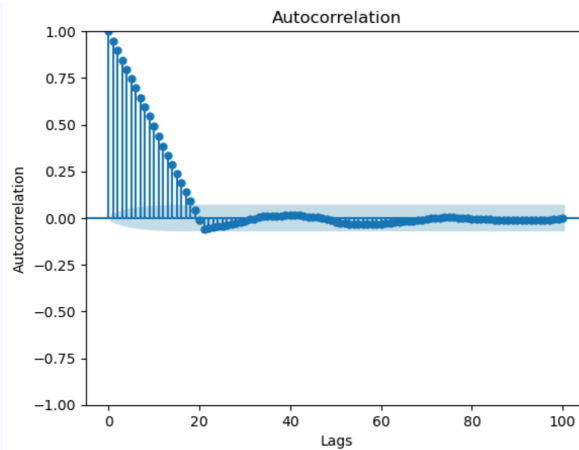
$$\gamma_v(s, t) = \begin{cases} \frac{3}{9} \sigma_w^2, & \text{if } |s - t| = 0 \\ \frac{2}{9} \sigma_w^2, & \text{if } |s - t| = 1 \\ \frac{1}{9} \sigma_w^2, & \text{if } |s - t| = 2 \\ 0, & \text{if } |s - t| > 2 \end{cases}$$

由此可见, autocovariance 的值仅取决于 s 和 t 之间的 lag 大小, 而不是 s 和 t 的具体取值



Python

```
1 win = 20
2 nt = 10000
3 w = white_noise(nt, var=5)
4 ma_data = moving_average(w, n=win)
5
6 plot_acf(ma_data[~np.isnan(ma_data)], lags=100);
7 plt.xlabel('Lags')
8 plt.ylabel('Autocorrelation')
```



⚠ Remark: 真实数据的 Moving Average 的 Autocovariance ∨

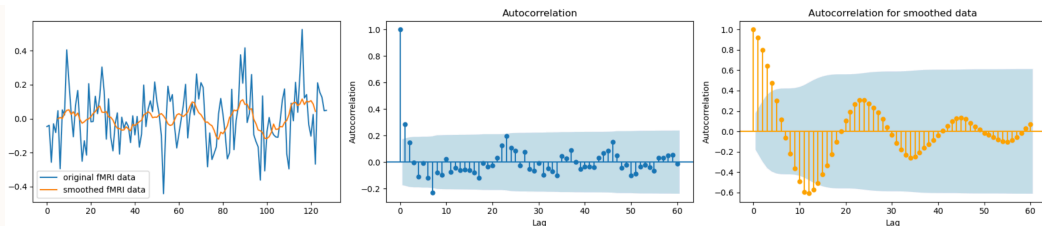
若对真实数据添加 moving average, 则会导致:

- 引入强 **autocovariance** 结构: 相邻的平滑值共享大量重叠观测, 因此会产生较大的正自相关, 即使原始数据相关性较弱
- **ACF** 在小 lag 处显著增大且衰减变慢: 平滑后序列的 ACF 通常呈现缓慢衰减或阻尼振荡形态
- 可能制造 "伪周期" 或 "伪长记忆" 现象: 在 ACF 中出现的振荡或长尾, 可能来自平滑操作本身, 而非数据真实机制



Python

```
1 import numpy as np
2 from matplotlib import pyplot as plt
3 import pandas as pd
4 from statsmodels.graphics.tsaplots import plot_acf
5 !pip install astsa
6 import astsa
7
8 fmri_data = astsa.load_fmri1()
9 region = 'thal2'
10 win = 11
11
12 # Apply the moving average function to the fMRI data:
13 ma_fmri = moving_average(fmri_data[region], n=win)
14
15
16 plt.figure(figsize=(18,4))
17 plt.subplot(1,3,1)
18 plt.plot(fmri_data[region], label='original fMRI data')
19 plt.plot(ma_fmri, label='smoothed fMRI data')
20 plt.legend()
21
22 plt.subplot(1,3,2)
23 plot_acf(fmri_data[region], lags=60, ax=plt.gca())
24 plt.gca().axis('tight')
25 plt.xlabel('Lag')
26 plt.ylabel('Autocorrelation')
27
28 plt.subplot(1,3,3)
29 plot_acf(ma_fmri[~np.isnan(ma_fmri)], lags=60, ax=plt.gca(), color='orange', vlines_kwargs=
30 {'color': 'orange'}, );
31 plt.gca().axis('tight')
32 plt.title('Autocorrelation for smoothed data')
33 plt.xlabel('Lag')
34 plt.ylabel('Autocorrelation')
35 plt.tight_layout()
36 plt.show()
```



1.3 Autocovariance of Random Walk

对于 random walk (with or without drift) $x_t = \delta t + \sum_{i=1}^t w_i$, 有

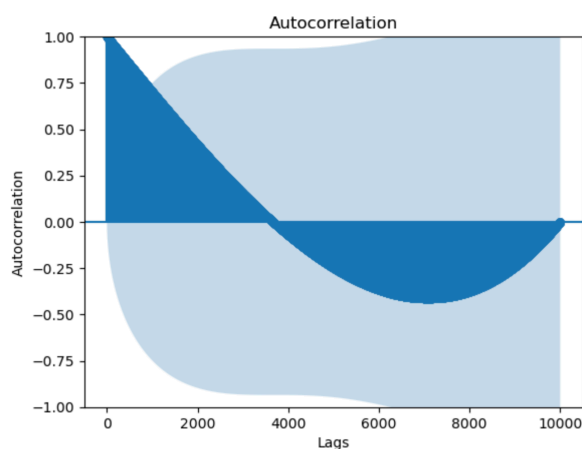
$$\begin{aligned}\gamma_x(s, t) &= \text{Cov}(x_s, x_t) \\ &= \text{Cov}\left(\delta s + \sum_{i=1}^s w_i, \delta t + \sum_{i=1}^t w_i\right) \\ &= \sigma^2 \min(s, t)\end{aligned}$$

此时 autocovariance 取决于 s 和 t 的取值



Python

```
1 nt = 10000
2 drift = 2
3 rwd = random_walk_drift(nt, drift=drift)
4 plot_acf(rwd, lags=len(rwd)-1);
5 plt.xlabel('Lags')
6 plt.ylabel('Autocorrelation')
```



Logic

我们可以将定义在同一 time series 上的 auto-covariance 的定义拓展到多个 time series 上, 即 cross-covariance

1.4 Definition: Cross-Covariance

对于两个 time series x_t 和 y_t 内的两点 s 和 t , 定义其之间的 covariance 为 **cross-covariance** γ_{xy} :

$$\gamma_{xy}(s, t) = \text{Cov}(x_s, y_t) = \mathbb{E}[(x_s - \mu_{xs})(y_t - \mu_{yt})]$$

Example: Same Time Series with Lag 2

考虑 $y_t = x_{t-2}$, 定义 $\gamma_{xy}(k)$ for lag k , 则

$$\gamma_{xy}(k) = \text{Cov}(x_t, y_{t+k}) = \text{Cov}(x_t, x_{t-2+k}) = \gamma_x(k-2)$$

当 $\text{lag } k = 2$ 时 γ_{xy} 最大

2 Correlation

Logic ▾

我们考虑 normalize auto-covariance 和 cross-covariance, 由此得到 auto-correlation 和 cross-correlation

2.1 Definition: Autocorrelation Function (ACF)

对于同一 time series x_t 内两点 s 和 t , 定义其 **autocorrelation function (ACF)** 为:

$$\rho(s, t) = \frac{\gamma(s, t)}{\sqrt{\gamma(s, s)\gamma(t, t)}} \in [-1, 1] \quad (\text{Cauchy-Schwarz inequality})$$

其衡量了 x_t 和 x_s 之间的 linear predictability

2.2 Definition: Cross-Correlation Function (CCF)

对于两个 time series x_t 和 y_t 内的两点 s 和 t , 定义其 **cross-correlation function (CCF)** 为:

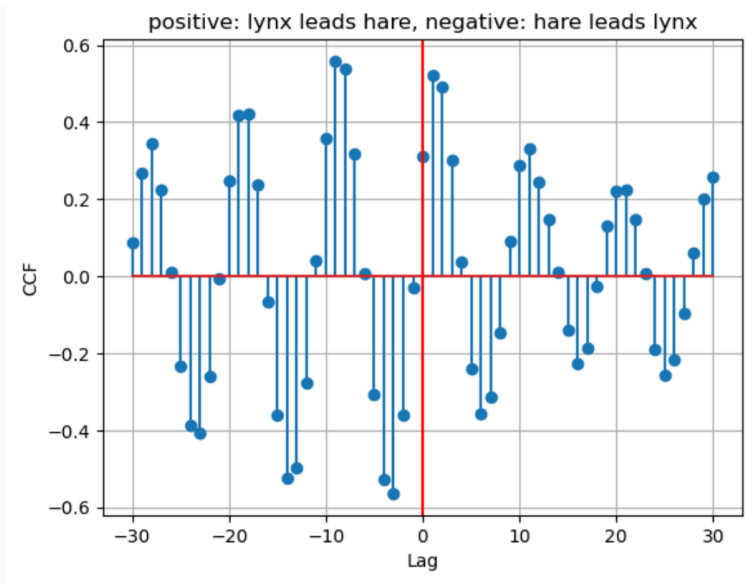
$$\rho_{xy}(s, t) = \frac{\gamma_{xy}(s, t)}{\sqrt{\gamma_x(s, s)\gamma_y(t, t)}} \in [-1, 1]$$

Example: 研究 lynx (猞猁) 和 hare (雪兔) 的种群变化关系 ▾



Python

```
1 hare_data = astsa.load_Hare()
2 lynx_data = astsa.load_Lynx()
3
4 x = lynx_data['Value'].values
5 y = hare_data['Value'].values
6
7 max_lag = 30 # Let's choose some value here explicitly
8
9 # Positive lags: lynx data leads hare data
10 ccf_xy = ccf(x, y, adjusted=False)[:max_lag+1]
11
12 # Negative lags: hare data leads lynx data
13 ccf_yx = ccf(y, x, adjusted=False)[:max_lag+1]
14
15 # Build symmetric lag axis
16 lags = np.arange(-max_lag, max_lag+1)
17
18 # Combine (exclude duplicate lag 0)
19 ccf_full = np.r_[ccf_yx[1:][::-1], ccf_xy]
20
21 plt.stem(lags, ccf_full)
22 plt.axvline(0, color='r')
23 plt.xlabel('Lag')
24 plt.ylabel('CCF')
25 plt.title('positive: lynx leads hare, negative: hare leads lynx');
26 plt.grid()
```



Logic ▾

我们已经在之前的例子中证明了, 对于 moving average, 其 autocovariance $\gamma_v(s, t)$ 仅取决于 s 和 t 之间的 lag
这点很重要, 且由此引出了 stationarity 的概念

3 Stationarity

3.1 Definition: Strictly Stationary

对于 time series x_t , 若所有的 collection of values 与其 time-shifted set 有相同的 probabilistic behavior (均值, 方差, 任意阶矩), 即

$$\{x_{t_1}, x_{t_2}, \dots, x_{t_k}\} \stackrel{d}{=} \{x_{t_1+h}, x_{t_2+h}, \dots, x_{t_k+h}\}$$

则称 time series x_t 为 **strictly stationary**

⚠ Remark ▾

Strictly stationary 的定义过于严格, 在大多数情况下都不成立

3.2 Definition: (Weakly) Stationary

对于 time series x_t (with finite variance), 若:

- Mean function 为 constant (不取决于 t)
- Autocovariance function $\gamma(s, t)$ 仅取决于 $|s - t|$ (lag)

则称 time series x_t 为 **(weakly) stationary**

⚠ Remark ▾

若 mean 和/或 autocovariance 随时间变化, 则 time series 不为 stationary

Logic ▾

由于 (weakly) stationary time series 的 autocovariance 仅取决于 lag, 我们可以优化 autocovariance 的定义

3.3 Definition: Stationary Time Series 的 Autocovariance

对于 (weakly) stationary time series x_t , 若 lag 为 $h = |s - t|$, 则其 autocovariance 可以被简化为:

$$\gamma_x(h) := \gamma_x(h, 0) = \gamma_x(s, t) = \text{Cov}(x_s, x_t) = \mathbb{E}[(x_t - \mu)(x_{t+h} - \mu)]$$

其有以下性质:

- **对称性:** $\gamma_x(h) = \gamma_x(-h)$
- **最大值:** $|\gamma_x(h)| \leq \gamma_x(0)$ ($\gamma_x(0)$ 为方差)

🔗 Proof: 最大值 ▾

由于 time series x_t 为 weakly stationary, 有 $\text{Var}[x_t] = \gamma_x(0), \forall t$, 因此由 Cauchy-Schwarz inequality, 有:

$$|\text{Cov}(x_t, x_{t+h})| \leq \sqrt{\text{Var}[x_t]\text{Var}[x_{t+h}]}$$

$$\Rightarrow |\gamma_x(h)| \leq \sqrt{\gamma_x(0)\gamma_x(0)}$$

$$\Rightarrow |\gamma_x(h)| \leq \gamma_x(0)$$

Logic ▾

对于 stationary time series, 我们可以在没有多个 repeated observations 的情况下 (无法直接估计 given time sample 的 variability), 估计 time series 的一些特征

3.4 估计 Stationary Time Series 的 Autocovariance

对于 (weakly) stationary time series, 若 lag 为 h , 则可以用单个 time series 估计 autocovariance:

$$\hat{\gamma}(h) = \frac{1}{n} \sum_{t=1}^{n-h} (x_{t+h} - \bar{x})(x_t - \bar{x})$$

其中, $\bar{x} = \frac{1}{n} \sum_{t=1}^n x_t$

且满足:

$$\hat{\gamma}(-h) = \hat{\gamma}(h), \quad \forall h = 0, 1, \dots, n-1$$

△ Remark: 为什么除以 n 而不是 $n-h$ ▾

虽然除以 n 后该 estimator 为 biased, 但有以下优势:

- 确保 estimators 构成的 autocovariance matrix 为半正定矩阵
- Estimator 的方差更低, 且仍为 consistent

Logic ▾

事实上, stationary 的定义可以拓展到两个/多个 time series 上

3.5 Definition: Joint (Weakly) Stationary (补充)

若 time series x_t 和 y_t 满足:

- x_t 和 y_t 均为 (weakly) stationary, 即
 - 均值 μ_x 和 μ_y 均为 constants
 - autocovariance γ_x 和 γ_y 仅取决于 lag h
- Cross-covariance $\gamma_{xy}(s, t)$ 仅取决于 lag $h := |s - t|$

Logic ▾

类似于单个 stationary time series, 对于 joint stationary time series, 我们可以在没有多个 repeated observations 的情况下 (无法直接估计 given time sample 的 variability), 估计 time series 的一些特征

3.6 估计两个 Stationary Time Series 的 Cross-Correlation

对于 jointly (weakly) stationary time series, 若 lag 为 h , 则可以估计 cross-correlation 为:

$$\hat{\rho}_{xy}(h) = \frac{\hat{\gamma}_{xy}(h)}{\sqrt{\hat{\gamma}_x(0)\hat{\gamma}_y(0)}}$$

Example

下图为 Southern Oscillation Index 和 Fish Recruitment

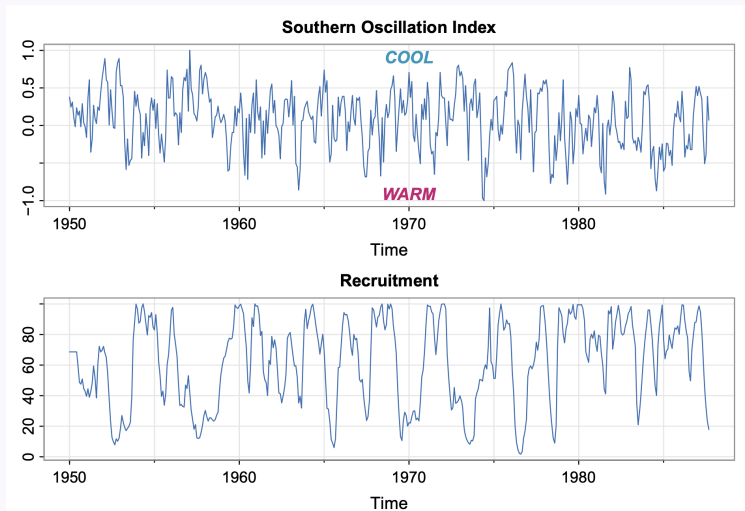


Fig. 1.5. Monthly SOI and Recruitment (estimated new fish), 1950–1987

从中可以得知:

- **周期性:** 两个序列均呈现周期性
- **负相关:** 视觉上两个序列呈现负相关

其 sample autocorrelation function (ACF) 和 sample cross-correlation function (CCF) 分别为:

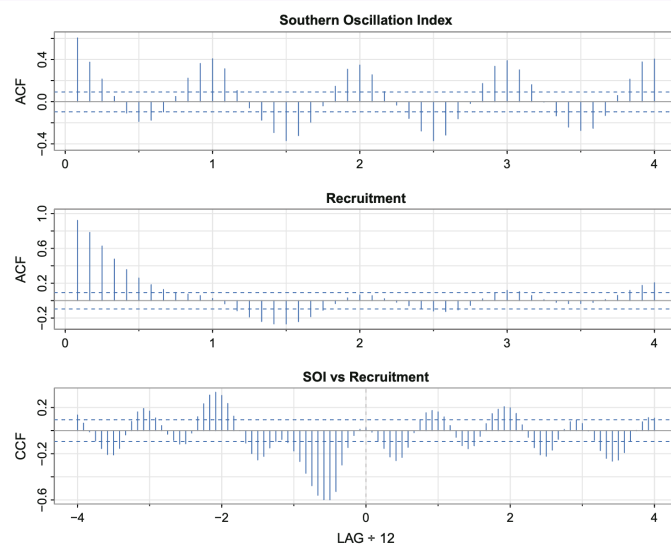


Fig. 1.18. Sample ACFs of the SOI series (top) and of the Recruitment series (middle), and the sample CCF of the two series (bottom); negative lags indicate SOI leads Recruitment. The lag axes are in terms of seasons (12 months)

从 ACF 图中可以得知:

- **SOI 存在年度周期性:** 在 lag 为整数 (每 12 个月) 处有高相关性
- **Recruitment 存在惯性:** lag 接近 0 时 autocovariance 很高, 初期的衰减较慢

从 CCF 图中可以得知:

- **约 1 年滞后的负向因果关联:** SOI 的变化领先于 Recruitment 约半年到一年